

JWALINSHAH

Fremont, CA · (408) 908-9659 · jwalinshah13@gmail.com

github.com/jwalin-shah · linkedin.com/in/jwalin-shah

AI Systems & Reliability Engineer · Agent Evaluation · Grounded Retrieval · Verification Infrastructure

Builds systems that constrain, observe, and verify probabilistic agents under real-world constraints.

PROFESSIONAL EXPERIENCE

tinyhumansai | Core Open-Source Contributor — OpenHuman

Mar 2026 – Present

- Owned the **life_capture** / **curated_memory** subsystem in the OpenHuman Rust core: on-device personal index ingesting iMessage, Gmail, Calendar, and Contacts into a structured local KB the agent reasons over.
- Shipped sequenced delivery (A1–A7): iMessage ingest controller, Composio→life_capture bridge for Gmail/Calendar, contact resolution via CNContactStore, EventKit Calendar/Reminders bridge, chronicle dispatcher + parser, session manager.
- 80+ upstream commits across memory, chat reliability, and security-policy hardening — productionizing personal-source ingest that previously lived as local prototypes.

Skill AI | Data Operations Lead (ML Systems & Reliability)

San Mateo, CA · Jul–Nov 2025

- Built data validation, collection, and evaluation systems across five robotic platforms; Python automation pipelines reduced operational overhead by 50% and improved downstream task success by 40%.
- Primary technical operator for 25+ live robot demos during \$1B Series C; maintained zero-failure execution under pressure.
- Hired 30+ operators end-to-end (sourcing through onboarding) and authored the training systems and runbooks that scaled the team across all five platforms.

Break the Web | AI Systems Engineer — LiveLM

Remote · 2026

- Built retrieval and evaluation infrastructure at Break the Web for LiveLM: ground LLMs in real-time knowledge with explicit hit / partial / stale / miss routing so the system abstains instead of inventing answers.
- Designed an MCP tool-use evaluation harness with eight independently measured stages; 71-question / 512-probe benchmark across serving paths.

PROJECTS & RESEARCH

Bridge — Reliable Coding-Agent Execution

Go · Seatbelt · Worktrees

- Spawn / verify / deliver pipeline with default-deny OS sandboxing and isolated worktrees; independent fresh-checkout verification so agents cannot certify their own success. Fail-closed gates; fixed a prior fail-open path; hardened adapters against catalogued failure modes.
- Catalogued 77 pipeline invariants with property tests plus TLA+ / Z3 / Lean verification layers — append-only evidence and independent checks instead of trusting the agent's self-report.

Knowledge Engine — Local Code & Axiom Graph

cocoindex · Neo4j · tldr

- Built a local Neo4j knowledge graph indexing axioms, source references, and code structure (cocoindex chunks + tldr CALLS) as the retrieval substrate Bridge consumes at spawn time.
- Certified catch-up indexing with parity gates: 3,460 non-empty source files matched 3,460 Neo4j Chunk nodes exactly — fail-closed when graph and disk disagree.

Sentient Arena (Cohort 0) — Grounded Reasoning Research

- Built a multi-agent LLM system for grounded financial reasoning, ranked top tier on a 246-task benchmark by replacing in-model arithmetic with structured retrieval and deterministic computation.
- Developed evaluation infrastructure across 15+ runs (3,600+ tasks); showed retrieval errors and tool selection — not model reasoning — drive most failures in grounded LLM tasks.

TECHNICAL SKILLS

Systems & Eval: Agent verification & fail-closed gates, grounded retrieval, tool-use eval harnesses, benchmark design, failure-mode analysis, hallucination / abstention measurement, on-device inference

Tools: Go, Rust, Python, TypeScript, FastAPI, SQLite + sqlite-vec, Neo4j, cocoindex, MLX, Seatbelt / sandboxing, TLA+ / Z3 / Lean (verification)

EDUCATION & LEADERSHIP

University of Texas at Dallas — B.S. Computer Science

GPA: 3.67

- UTD Ultimate Frisbee — Captain: led 30-member competitive program through recruitment, budgeting, and tournament logistics.
- UTD Chess Club — USCF 1832 (91st percentile); coached 50+ students in positional analysis.